

THE GEOGRAPHY OF BORROWING SIZE: EXPLORING SPATIAL DISTRIBUTIONS FOR GERMAN URBAN REGIONS

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ABSTRACT

This paper contributes to the discussion on borrowing size effects. According to this concept, smaller cities that are part of larger functional urban regions can utilise metropolitan functions and economic externalities, thereby boosting their regional performance. This is conceptualised in the four-dimension scheme of Meijers and Burger. Consequently, this paper analytically explores and operationalises the borrowed size concept and reveals insights into the relation and spatial distribution of four types of effects: borrowed size, borrowed performance, borrowed function and agglomeration shadow. Research on these spatial effects is applied on a small-scale spatial level and integrates all municipalities in German urban regions. We find different geographies of effects among the four-dimensions of borrowed size. City size and centrality degree show a significant influence on positioning in the four borrowed size effect types.

Key words: Borrowed size, borrowed function, borrowed performance, urban growth, metropolitan functions, urban regions

INTRODUCTION

The current urban development in Europe can be characterised by two important trends. In many cases, the capacity to attract high-order or metropolitan functions (Friedmann 1986; Sassen 1991) and economic development are decoupled from urban size effects (Camagni & Capello 2015), which contradicts standard agglomeration theories (Camagni *et al.* 2016). There is no clear relationship between urban scale or density and urban productivity (Cox & Longlands 2016; McCann 2016). In addition, the spatial patterns of regional urban growth in European regions are very heterogeneous (Dijkstra *et al.* 2013) in their extent and direction and are influenced by path dependencies, different factor endowments and specific interregional dependencies within and

among urban regions (Camagni *et al.* 2016). For regional policies and policy measures, it is important to be able to identify and classify the phase and spatial shape of regional developments to adapt the right kind of urban policy (Dijkstra *et al.* 2013), as, for instance, monocentric urban growth necessitates other concepts of infrastructure plans than in more polycentric city development.

In explaining these trends and developments, the contemporary literature highlights a complex understanding of urban and regional growth, which is rooted in different disciplinary and methodological backgrounds. One interpretation of agglomeration economies and urban dynamics is connected to the concept of ‘borrowing size’ (introduced by Alonso 1973; and reintroduced by Phelps *et al.* 2001 & Phelps 2004;

and currently discussed by Burger *et al.* 2015; Meijers *et al.* 2016; Meijers & Burger 2017). The theoretical underpinning of this concept discusses the functional enrichment of medium and small cities that are part of larger functional urban regions, where such cities are able to utilise metropolitan urban functions and networks, thereby boosting their regional performance (which can be economic or demographic). In contrast, cities that fail to benefit from the functions and power of bigger neighbours may experience a 'backwash' or 'agglomeration shadow' effect. Here, a major urban core is dominant in its metropolitan functions and economic development compared to its hinterland (Cardoso & Meijers 2016). The conceptualisation of the borrowing size dimensions by Meijers and Burger (2017) provides a basis for its operationalisation. The contemporary empirical findings on the borrowed size concept reveal a gap in urban research regarding the empirical operationalisation and analysis of the spatial distribution and classification of cities in urban regions on a small-scale level using a solid set of indicators and a consistent method.

In this paper, we briefly discuss the current state of the art for the concept of borrowed size, focusing on recent empirical findings and advances in logical classifications (second section). The following section introduces our study area – a set of large, medium and small cities within the functional German urban regions – and presents the analytical methods and data used to empirically explore the spatial heterogeneity of borrowed size effects (third section). For this, we analyse the spatial patterns of borrowing size effects and the interrelations among functions, performance and agglomeration shadow in the cities of the observed German urban regions (fourth section). The results of this empirical study are discussed in the fifth section.

THEORETICAL FOUNDATION

Concept and origin of 'borrowed size' – The origin of the borrowed size concept was introduced by Alonso (1973, p. 200), who proposed

the idea that a smaller city 'exhibits some of the characteristics of a larger one if it is near other population concentrations'. This analytical concept focuses on the understanding that people and firms in smaller settlements (low concentration and low rents) retain advantages from their locality and are able to exploit advantages from larger nearby settlements (e.g. access to diverse market and services, infrastructure, urban amenities) and avoid agglomeration costs (Camagni *et al.* 2016; Meijers & Burger 2017).

An investigation of economic developments in Greater London by Phelps *et al.* (2001) tested the concept and revealed evidence of borrowing size effects in edge cities, whereby small firms derived advantages from nearby larger urban areas and not from their own local economy. The concept of borrowed size refers to the process of suburban growth interlinkages, to discussions on edge cities (Phelps 1998) and new economic subcentres (Phelps 2004; Krehl 2015), which are integrated into polycentric urban forms (Phelps 2004). 'It derives from the tension between forces of agglomeration, on the one hand, and those of decentralisation on the other hand' (Phelps 2004, p. 981).

Currently, there is a variety of papers (Burger *et al.* 2015; Malý 2016; Meijers & Burger 2017) dealing with the question why some smaller cities in proximity to larger cities have special prerequisites for relatively high economic and population growth. Here, the borrowed size concept is used to explain the need for geographical proximity and functional connectivity to foster economic advantages and a thriving regional economy (Burger *et al.* 2015). In general, results indicate that a place borrows size when it hosts more metropolitan functions than its own size could normally support.

Locations in proximity to a larger centre within an urban region can also be related to negative borrowing size effects, the so called 'agglomeration shadow' effects. This term is applied in new economic geography (Krugman 1991), which states that 'growth near concentrations of firms will be limited by competition effects. Positioning within the "shadow" is not profitable for firms' (Meijers & Burger 2017, p. 274). That entails a low-risk for agglomeration shadow effects in isolated cities compared

to cities that are nearby large cities. This is in line with the ideas from central place theory (Christaller 1980) and urban systems theory (Berry 1964), where smaller cities have fewer central place functions or consumer amenities than isolated cities of similar size (Burger *et al.* 2015; Cardoso & Meijers 2016). In addition, there are also major overlaps to the regional concept of spread and backwash effects, conceptualised by Myrdal (1957). It was used to investigate and understand the relationship of growth and decline between urban and rural areas (Henry *et al.* 1997). Spread effects occur from a centre to its hinterland, causing economic and population growth. This is in line with the concept of borrowed size effects for cities nearby large cities in which a city profits from urban functions (e.g. infrastructure, public services) or in economic performance in terms of population and employment growth (Veneri & Ruiz 2016). Backwash effects can be described as a process in which the hinterland allocates resources to centres. These negative effects are very similar to the agglomeration shadow effect.

Studies and operationalisation of borrowing size, function and performance—Theoretical and empirical studies build upon and enrich the concept of borrowed size (Polese & Shearmur 2006; Camagni & Capello 2015). They are discordant regarding where, in which regions, and in which cities borrowing size effects occur. Brezzi and Veneri (2015) note that cities near large agglomerations seem to have no significant effect on regional performance (GDP per capita) in multi or polycentric areas, and, even if cities are about the same size, the regional performance is negative. This evidence corresponds to outcomes in north-west Europe (Burger *et al.* 2015) and the Netherlands (Meijers 2008), in which larger cities with high-order functions spread a shadow over the smaller surrounding centres. This results in places that are in the shadow of larger cities that possess fewer functions than one would expect from their size.

Other empirical studies provide an indication of borrowed size effects; smaller cities in large urban regions show positive employment

(Polese & Shearmur 2006) and population (Partridge *et al.* 2007) growth rates. The magnitude of effects depends on the utilisation of high-order urban functions and networks such as urban infrastructure, density to external linkages, mobility factors and education (Polese & Shearmur 2006; Camagni *et al.* 2015). Veneri and Ruiz (2016) confirm that population growth depends on the significance of the proximity to the closest main centres – increasing the distance from a large centre increases transportation costs and reduces population growth, benefits and the use of technology. Some cities, primarily those in long distance to urban centres or those that are small in terms of economic size or old in demographic structure, are not able to benefit from growth effects and are affected by backwash effects/agglomeration shadow effects (Partridge *et al.* 2007).

Differences between classes of urban regions cannot be explained by merely considering agglomeration economies and diseconomies. The benefits of spatially clustered cities are related to their position relative to another city, so borrowing size is linked to the interactions that generate agglomeration economies (Polese & Shearmur 2006). Travelling distance (km or time) is one indicator for access to agglomeration externalities. Partridge *et al.* (2007) constitutes that, with every kilometre increase in distance from the small city to the large one, cities experience less population growth. A travel time of one hour allows firms located in smaller cities to access specialised labour market and informational external economies (Phelps *et al.* 2001). The travel time for accessibility or connectivity is not the only crucial factor that generates spatial effects for the urban hinterland. According to Hesse (2016), the co-operative relationships and exchange of information are important in addition to the physical distance. Cities embedded in specific national or international networks can benefit from borrowed size effects – ‘borrowed size is less a product of distance or access than it is of true interaction’ (Meijers & Burger 2017, p. 288). Hence, smaller cities can substitute their lack of urban mass or city size by being integrated in

a (inter)national network (Meijers *et al.* 2016). Larger cities with a certain size and presence of metropolitan functions play a crucial role for urbanisation economies because they have access to a larger (inter) national market than smaller cities. Burger *et al.* (2015) note that size is the most important factor explaining the presence of high-order or metropolitan functions because a critical mass is crucial and necessary for the existence and gathering of such urban amenities.

High-order or metropolitan functions are ‘characterised by higher thresholds for the level of appearance in the city (in terms of urban population)’ (Capello & Camagni 2000, p. 1483) and are sometimes used in terms of high-level occupations, particularly gaining a high percentage of the financial and corporate service sub-sectors (Camagni *et al.* 2016; Hesse 2016) or service sector employment and office and retail floor space (Phelps 1998). Another approach in this regard addresses functional characteristics and indicators that can be constructed in functional urban regions (BBSR 2011). This method is used to identify and select high-level functions of cities using concepts and classifications covering wide-ranging: (i) decision-making and control functions by public and private sectors; (ii) innovation and competition functions; (iii)

gateway functions; and (iv) symbolic functions (Behrendt & Kruse 2001; Blotevogel & Danielzyk 2009). These functions can be viewed as an expression of geographical power and relate to the hierarchical understanding of urban regions as part of a central place system (Meijers 2007).

That leads to the conclusion that borrowing size effects are stronger where physical proximity and accessibility to large cities can be exploited, which underlines the argument for better performance of cities in more polycentric urban regions due to interactions among actors sharing agglomeration externalities. However, analysis of spatial and functional differentiations within urban systems are scarce in terms of their view on urban functions and urban growth interdependencies (Cardoso & Meijers 2016).

Redefining ‘borrowed size’ – Recent studies by Meijers and Burger (2017) and Meijers *et al.* (2016) redefined and stretched the concept of borrowed size along several dimensions in terms of scale and scope. The authors propose a distinction between two dimensions of borrowing size, namely, ‘borrowed performance’ and ‘borrowed functions’ (see Figure 1). Borrowed performance is defined by performance that is better than expected given

Dimensions of borrowed size		Connection size ↔ function	
		Less functions than expected given size	More functions than expected given size
		Agglomeration shadow	Borrowed function
Connection size ↔ performance	Performs less than expected given size		
	Performs better than expected given size	Borrowed performance	Borrowed size (function & performance)

Source: Meijers and Burger (2017).

Figure 1. *Dimensions of borrowed size.*

the size and refers to advantages derived from a pooled and diversified labour market and population spill-overs. Borrowed functions are defined as more functions than expected given the size and can be linked to accumulations of high-order/metropolitan functions, such as artefacts, activities, amenities. If both processes occur simultaneously, the authors define it as the 'borrowing size' dimension (Meijers & Burger 2017).

The question of agglomeration economies of immobile high-order functions such as universities, airports or infrastructure in urban centres is closely correlated to the spatial distribution of borrowed size effects. With access to functions of large cities through connectivity (true interaction), smaller cities in the surrounding area can borrow central or metropolitan functions from the city, which may manifest in higher population growth or higher incomes but can unfortunately prevent surrounding cities from having the ability to perform important functions themselves:

Whereas growth potentials in terms of urban functions are limited in the surrounding areas of large dominant cities, this does not necessarily mean that population growth is equally restricted (Cardoso & Meijers 2016, p. 1002).

Meijers and Burger (2017) conclude that smaller cities support larger cities to maintain more metropolitan functions, so the borrowing function occurs more frequently within larger cities, while smaller cities are typically the borrowing performance type. Larger cities are likely to cast a functional agglomeration shadow over the entire urban region by concentrating many metropolitan functions in its core.

Research gap – In this contribution, we investigate the possibilities to operationalise the (stretched) concept of borrowed size. We investigate three research questions that are directly derived from the current state of the art in borrowed size research:

(1) How can borrowed size categories be measured statistically?

The first objective is to analytically explore and operationalise the concept of

the borrowed size. We adopt the redefined concept with its four-dimensions (Meijers & Burger 2017) and explore the relation among borrowed size, performance, function and agglomeration shadows. Therefore, it is necessary to translate the theoretical and analytical matrix into measurable indicators to categorise cities within the German urban regions in terms of their borrowed size dimension.

(2) To what extent are cities in German urban regions affected by spatial effects of borrowed size?

We assume that smaller cities in urban regions are affected by borrowed size effects and expect regions that are affiliated with a polycentric urban region enjoy borrowed performance or borrowed function effects. More central cities that are predominant to their surroundings should be characterised by borrowed size effects and cast an agglomeration shadow on their functional hinterland. The investigation of these spatial distributions is applied on a small scale and integrates all large, medium and small municipalities in German urban regions to determine which cities are affected by various borrowed size effects.

(3) Do city size and the spatial structure have different effects on borrowing function and borrowing performance?

Due to the different spatial structure of German urban regions, we expect different effects along the four-dimensions of borrowed size. We want to find out whether these effects in large cities behave differently *vis-à-vis* smaller and medium cities.

RESEARCH DESIGN

Study region – Our spatial reference in this paper is the German urban system. Unlike the French or English urban systems, the German urban system does not contain a large primate city. Instead, it has a polycentric structure of 10–12 leading core cities with significant economic, political and cultural functions (Blotevogel 2000), which makes it a particularly interesting example to study borrowed size effects. Due to Germany's polycentric structure and its historical development – espe-

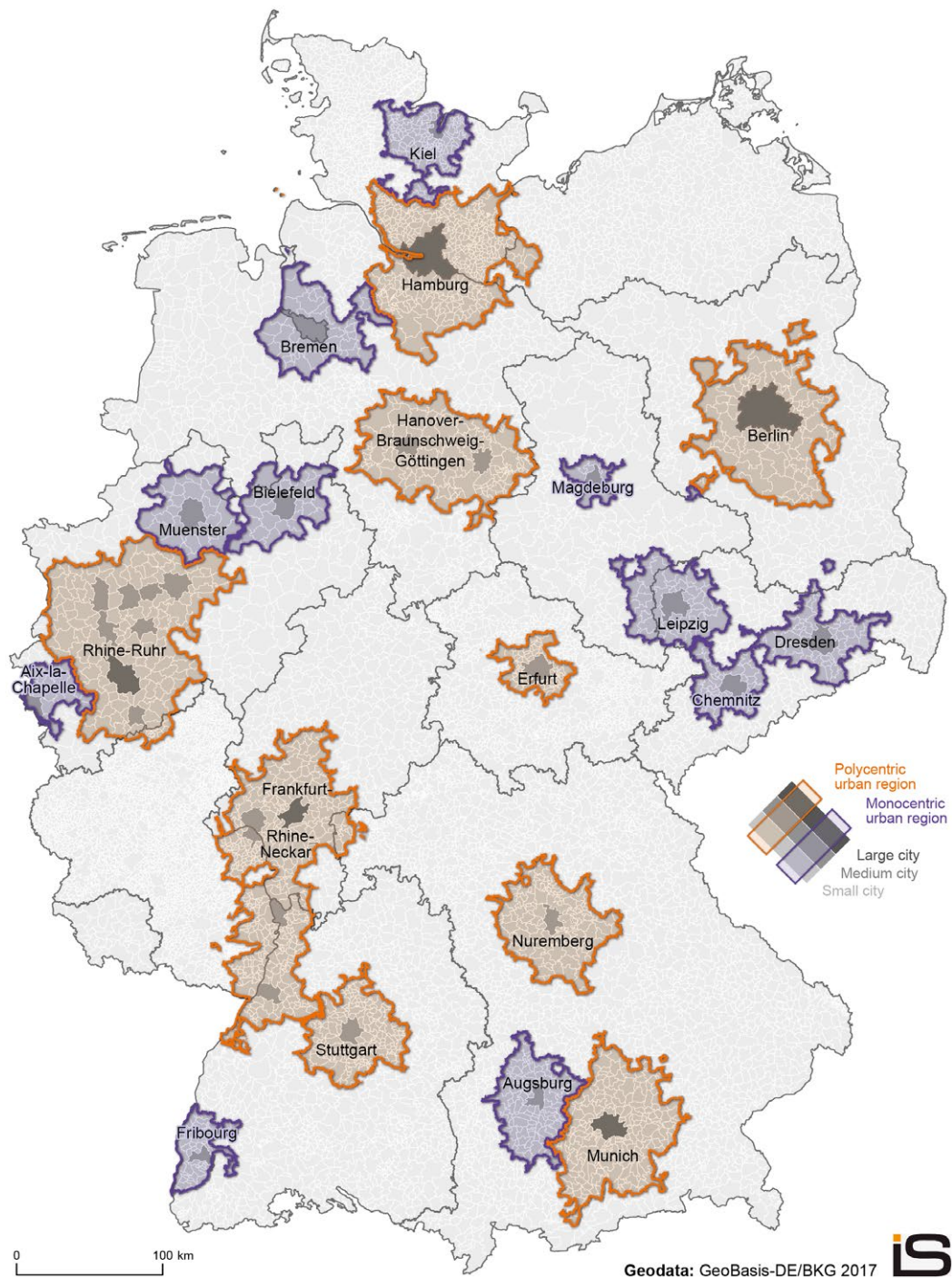
cially since reunification – the German urban system constitutes an interesting case study because, within the urban system, there is a wide range of urban regions with different structural and functional conditions. The origin of this structure dates back to the Middle Ages and continues to be reflected in the federal structure of Germany's political and administrative system (Blotevogel & Hommel 1980). The division of the urban structure was a result of the Second World War and its political aftermath; political-administrative and business functions were removed from West Berlin and relocated to a number of major regional centres in West Germany. As a consequence, West Germany's urban system developed a complex polycentric structure (Krätke 2002). In contrast, East Germany continued to be dominated by (East) Berlin, thereby acquiring a monocentric structure (Prigge & Schwarzer 2006). Since reunification in 1990, the system has undergone several dynamic changes. Berlin's status as capital city in 1999 and the following relocation of the government led to a repositioning of German cities, however Berlin is not comparable to Paris or London.

We use a specific functional delineation of urban regions (Siedentop & Kaup 2017). The spatial references are the German municipalities, so the analysis is done on a small scale. In the first step, we define core cities with at least 200,000 inhabitants and 100,000 employees/jobs (within social insurance). In accordance with these criteria, we identified 32 core cities in the German urban system. Within this group of cities, there are five major large cities with more than 600,000 inhabitants: Berlin, Munich, Hamburg, Frankfurt and Cologne. The remaining 27 core cities are defined as medium cities. It is assumed that each of these core cities plays an important functional role for its urban hinterland, providing higher-level central places such as economic hubs and employment centres. The functional hinterland of the 32 core cities is calculated by a network analysis, routing from the core city to each municipality focal point (in the hinterland) with a defined threshold for real car travel times. The commuting area of a core city was based on a gravitation-curve derived from the number

of employees. The core city with the highest employment centralisation had a 60-minute commuting buffer (Berlin), while the lowest was 30 minutes (Erfurt) (see Figure 2). This delineation leads to overlapping urban regions, for example, in Duisburg, Essen and Dortmund. Thus, we use a disjunct delineation of urban regions that sorts the municipalities to their individual nearest core (in travel time). This resulted in a group of 20 urban regions, which can be differentiated into monocentric (with one core city) and polycentric (at least two core cities located in the functional urban region) urban regions with one or more core cities that are non-overlapping areas. In extension, all municipalities that show metropolitan functions (according to our indicator) and are neither large in type nor classified as a city regional core, are classified as small cities.

Operationalisation of the stretched concept of borrowed size – As discussed in the introduction of the concept, there is a need to define performance as well as metropolitan function variables. In addition, the conceptualisation of the four-field classification relies heavily on the idea that regions perform better or have more functions than you would expect given their size. To enable classification of regions, there is a need for readily definable variables and a concept for defining whether regions do better in a specific variable than one would expect from their size. In the following two sections, we introduce one variable for metropolitan functions and another as a measure for regional performance. In the next section, we discuss how expectations are incorporated into the concept and how we applied it to our sample.

Metropolitan functions – The operationalisation of borrowed functions involves the question of how to enumerate the metropolitan characteristics of a large city to compare them with other cities. A set of indicators was designed to operationalise and quantify metropolitanity. The selection of suitable indicators is determined by whether the pointers reflect metropolitan



Source: data from The Regional Database Germany. Cartography: Jutta Rönsch.

Figure 2. German urban regions.

characteristics, namely, whether they conjure up a metropolitan flair or importance (Krätke 2003). Characteristics that purely refer to a city's structure or trivial size effects such as population or population density are not sufficient indicators. The presence of high-level functions is frequently connected to urban size, but not always. For example, a city such as Zurich, with only 350,000 inhabitants, is specialised in international finance like New York and Tokyo (Capello 1998). Furthermore, the choice of an appropriate indicator depends on a comprehensive data structure for the study region (counties and cities in Germany), the comparability of data sources among time periods and objective and reliable statistics (Volgmann 2014; Growe & Volgmann 2016). Our dataset consists of 47 localised functional attribute indicators for 2008–2010 (see Volgmann 2014; and Table 1). To simultaneously detect and compare a number of metropolitan facets, the raw data were z-standardised and a metropolitan index for all 47 indicators was constructed from the various indicators (a static indicator) using additive linking. Due to the analysis on the municipality level, the metropolitan index of the counties was transferred to the largest municipality belonging to a county. Each urban municipality has its own index.

Urban performance – Although a variety of papers describes concepts for measuring metropolitan functions, there is a lack of empirical answers to the question of how to measure urban performance. In their paper on stretching the concept of borrowed size, Meijers and Burger (2017) describe the general dimensions of performance but do not operationalise the idea in detail. We pair urban performance with metropolitan functions by focusing on a readily available basic economic variable of municipalities: the number of jobs (within social insurance), which is accessible via public regional data sources (regionalstatistik.de). In doing so, we narrow the view on performance to a purely economic point of view, which may not be optimal. For instance, other authors discuss adding information on demographic or income indicators, for example Cardoso and

Meijers (2016). Due to a lack of income data on such a local scale and the intended focus on the demand for labour, we suggest using employment data, as it is available on small scales as well as for long time horizons.

In addition to the general choice of data to measure performance, it is important to discuss the indicator used to measure the relative performance (a dynamic indicator) of the urban regions. In our view, it is misleading to focus on the absolute change in employment or the pure percentage change in employment in this case. In this research, 'performance' and the shift or relative importance of cities are of utmost importance. When examining absolute changes in employment, cities with an already large employment base tend to have above-average changes between years. Additionally, when examining relative changes, small absolute changes in cities with a small employment base may have relatively high percentage changes. Therefore, our analysis uses the employment shares of cities, which is often used when comparing the economic performance of cities (Dijkstra *et al.* 2013).

Following this logic, we calculate the employment share of each city with a metropolitan function out of the sum of employment of these cities. We use a subset of 60 municipalities in this case. Then, we compute the changes in those shares between 2008 and 2015. We consider cities that are able to increase their share as performing relatively well. Cities that lose relative importance perform relatively poorly.

Expectations 'given size' – The final step in operationalising the concept of borrowed size is to define how it can empirically be employed to measure the expected performance and functions. This enables us to assign regions along the dimensional scope of the concept.

Meijers and Burger (2017) suggest using a parsimonious regression approach for metropolitan function endowment for regions. In their approach, they regressed size on an index of metropolitan functions and used the residuals of that regression to identify how many metropolitan functions a region has. In this empirical logic, a region with a positive residual can be classified as providing more

Table 1. *Forty seven localised functional attribute indicators.*

Control function	Innovation function	Gateway function	Symbolic function
Business and finance	Research/development in the private sector	Transportation	Media and cultural economy
headcount of the 500 largest companies	- locations of the 100 most innovative companies in Germany	- flight activity at international airports	- locations of private and public broadcasting companies
- turnover of the 500 largest companies	- number of engineers employed	- passenger volumes at international airports	- film studios
- total assets of the 50 largest banks	- employees working in business-oriented services	- ICE/TGV stations	- the 100 largest book publishers
- gross income of the 30 largest insurance companies	- employees with an academic degree		- national newspaper publishers
- stock exchange locations	- patents filed by companies		- Internet domains
- turnover of the top 30 retail food companies			- employees working in cultural occupations
Policy	Science and research	Market volume	Arts, culture and architecture
- locations of federal ministries	- locations of DFG special research areas	- transshipment volumes in seaports	- locations of buildings by famous architects
- number of employees working for the Federal State and the Länder	- research locations belonging to the Helmholtz Community	- transshipment volumes in inland ports	- locations of Germany's 25 highest buildings
- locations of courts	- locations of Max-Planck, Fraunhofer, Leibniz, Academies and Leopoldina	- turnover of the top 100 logistic companies (in € million)	- location of major urban development competitions
- EU/UN institutions	- university locations	- air freight volumes at airports	- opera audiences
- embassies and consulates	- media units in science-oriented general and university libraries	- trade fair/exhibition locations – exhibition floorspace (> 100,000 sqm)	- theatre audiences
- associations	- patents filed by research institutes		- concert audiences
- development organisations	- employees working in research		- capacities of the 20 largest football stadia
- foundations			- hotel guests

Source: Volgmann (2014).

metropolitan functions than could be expected using the size as an indicator and vice versa.

In our approach, we also use this general logic of regression residuals as an indicator for more or less endowment 'given the size'. Importantly, we extend this logic in two aspects. We introduce our performance measure as the second pillar to address the full scope of borrowed size dimensions. We use the residuals of two regressions, which are constructed following general recommendations

in economic geography literature (Combes & Gobillon 2015). Researchers suggest using information on employment or population to measure the size of a local economy. They strongly advise using information on density when analysing regional entities (Ciccone & Hall 1996). Accordingly, we use parsimonious regressions, as suggested by Meijers and Burger (2017), and add density as an explanatory variable, which is the second conceptual amendment:

MF = POP + POP.DENSITY, (1)

EMP.SHARE = EMP + EMP.DENSITY, (2)

where MF is the metropolitan index, POP population, and POP.DENSITY is the population divided by the settlement area. All values are from 2008. We opted for the use of static indicators because the metropolitan functions do not notably differ between 2008 and 2015. In contrast, in the second regression, the change of the employment share of a municipality between 2008 and 2015 (EMP.SHARE) is explained by the stock of employment in 2008 (EMP) and the employers per settlement area in 2008 (EMP.DENSITY). In the second case, we want to explain a change in employment share to reflect the concept of relative urban performance.

As shown in Table 2, we used the residuals of both OLS regressions to classify the 60 municipalities with metropolitan functions along the four dimensions of borrowed size. For the metropolitan index, we obtained similar explanatory power (adj. $R^2 = 0.86$) as the OLS models reported in Meijers and Burger (2017), whereas the employment share calculations performed worse but were still acceptable, with an adjusted R^2 of 0.51.

ANALYSIS AND EMPIRICAL FINDINGS

This section presents a comprehensive picture of the spatial distribution of metropolitan functions and the economic performance in the cities of German urban regions. The spatial distribution of the overall metropolitan index in 2008–2010 is shown in Figure 3 for 60 municipalities in German urban regions (above average value). This reveals a hierarchical system dominated by – roughly – five

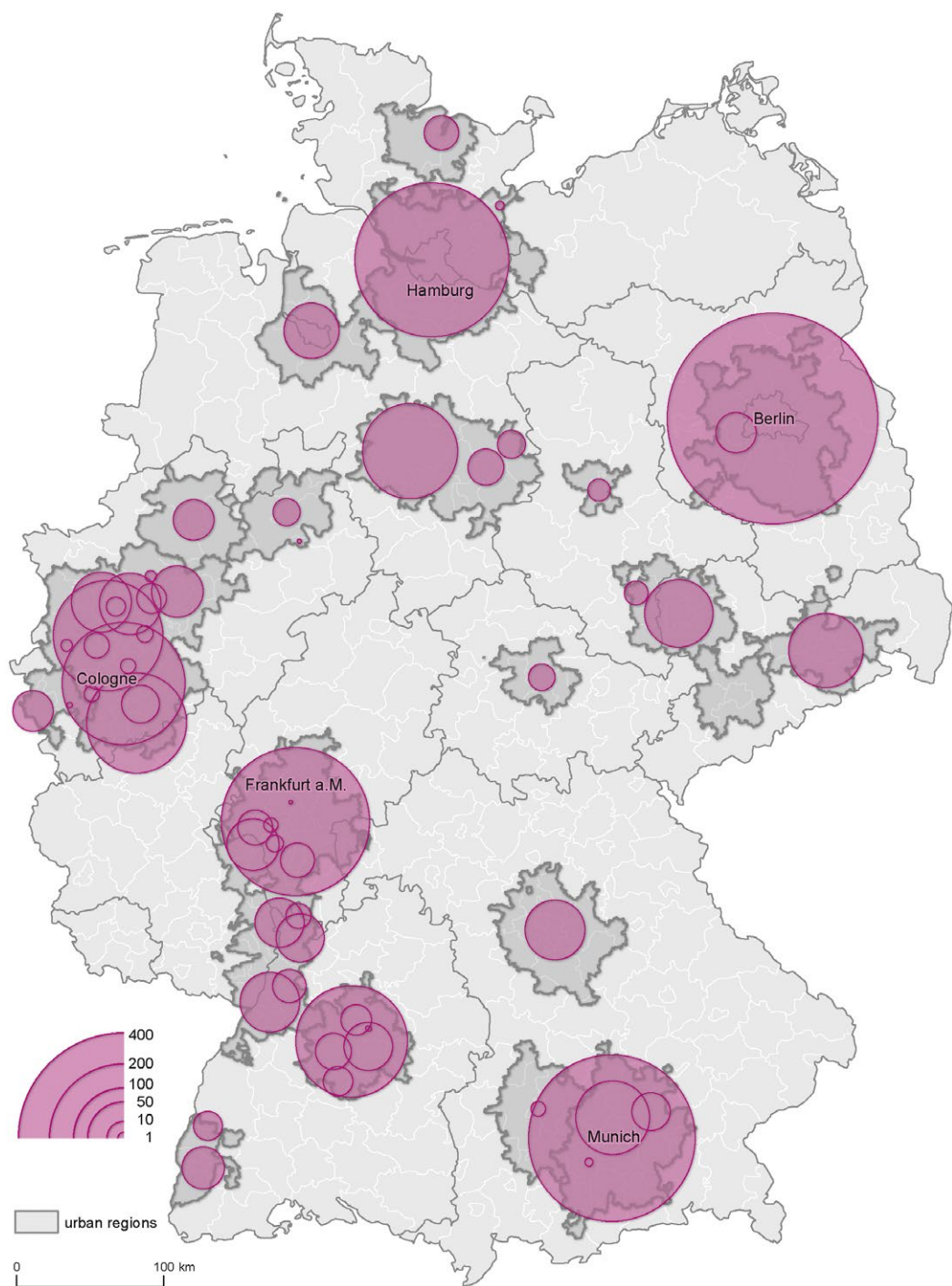
core cities, with Berlin (414.76) and Munich (260.54) outperforming other German cities, emphasising their dominant function. Together with Hamburg (221.80), Frankfurt (206.12) and Cologne (141.03), they are the leading cities in the German urban system. This corresponds to the theory-based assumption of understanding metropolitan functions in the sense of central place theory. The gains of Berlin trace back to the relocation of the government to Berlin. Members of Parliament and their offices, embassies, representative offices of the federal states, political parties, associations, foundations, interest groups and media representations moved to Berlin. The capital of Berlin thus benefited from the relocation of public and cultural institutions and from funding and infrastructure investments and became an attractive destination for visitors and tourists. Together with the medium sized cities these five large cities build the central cores of the urban regions of the following analysis, where borrowed effects are investigated. Because of the polycentric structure in the German urban system, we expect different spatial interdependencies – some nearby cities may benefit from borrowed function and borrowed performance but some cities may be affected by agglomeration shadows.

Figure 4 depicts the changes in employment shares, which is, in our definition, the ‘performance’ of regions for all municipalities in the functional urban regions analysed. The 60 cities with a metropolitan index value are highlighted. Notably, the large cities in the German urban system are also those that perform quite well in relation to other municipalities. Berlin, Munich, Hamburg and Cologne led in increasing their share in employment from 2008 to 2015. Interestingly, Leipzig, a medium city, is one of the more thriving

Table 2. Operationalisation of the four dimensions of borrowing size.

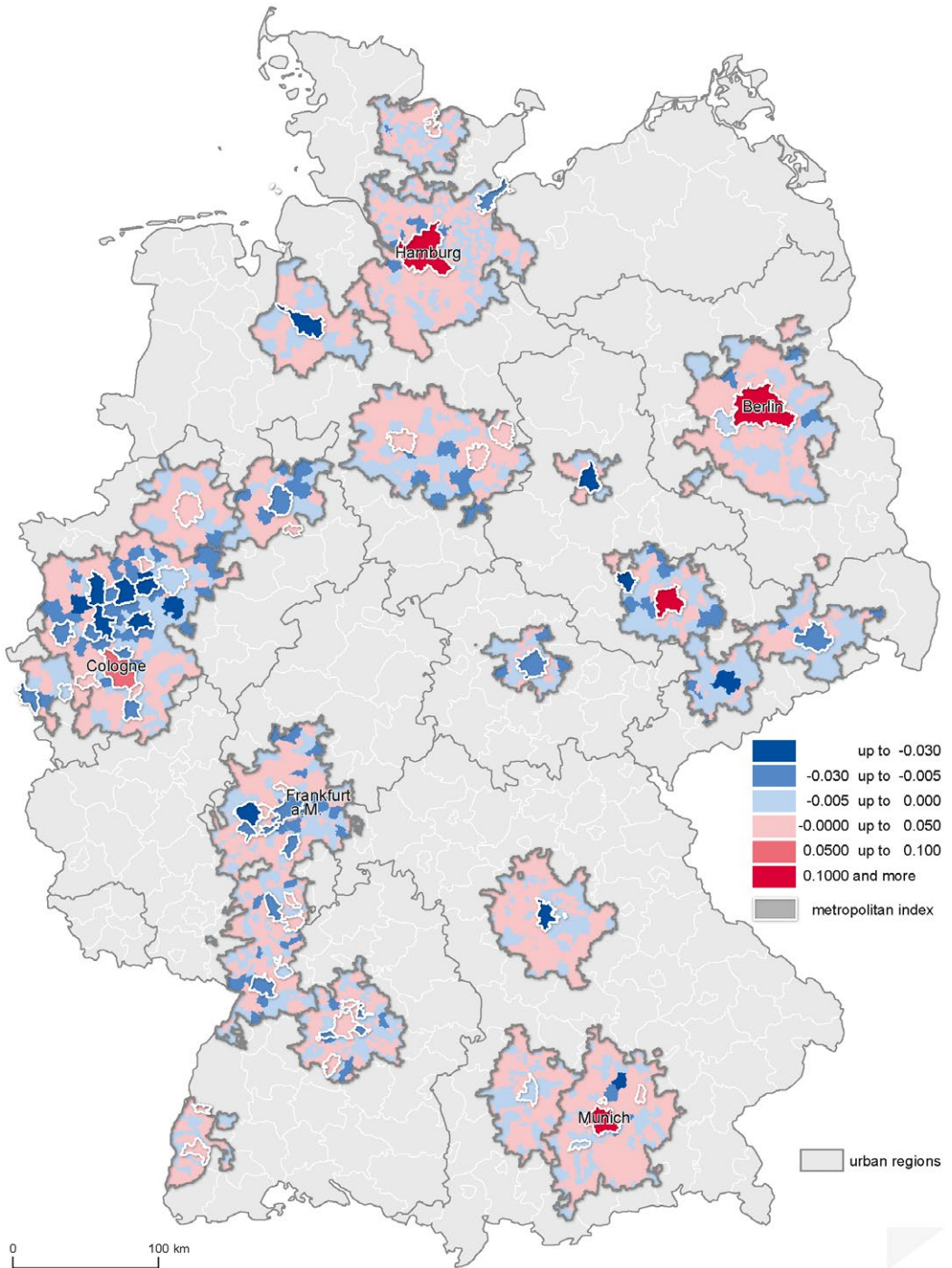
		Regression on metropolitan index	
		Negative residuals	Positive residuals
Regression on shift in employment share	Negative residuals	Agglomeration Shadow (20 municipalities)	Borrowed Functions (7 municipalities)
	Positive residuals	Borrowed Performance (16 municipalities)	Borrowed Size (17 municipalities)

Source: own illustration based on Meijers and Burger (2017).



Source: own calculations. Cartography: Jutta Rönsch.

Figure 3. Metropolitan index in cities of German urban regions (2008–2010).



Source: data from The Regional Database Germany. Cartography: Jutta Rönsch.

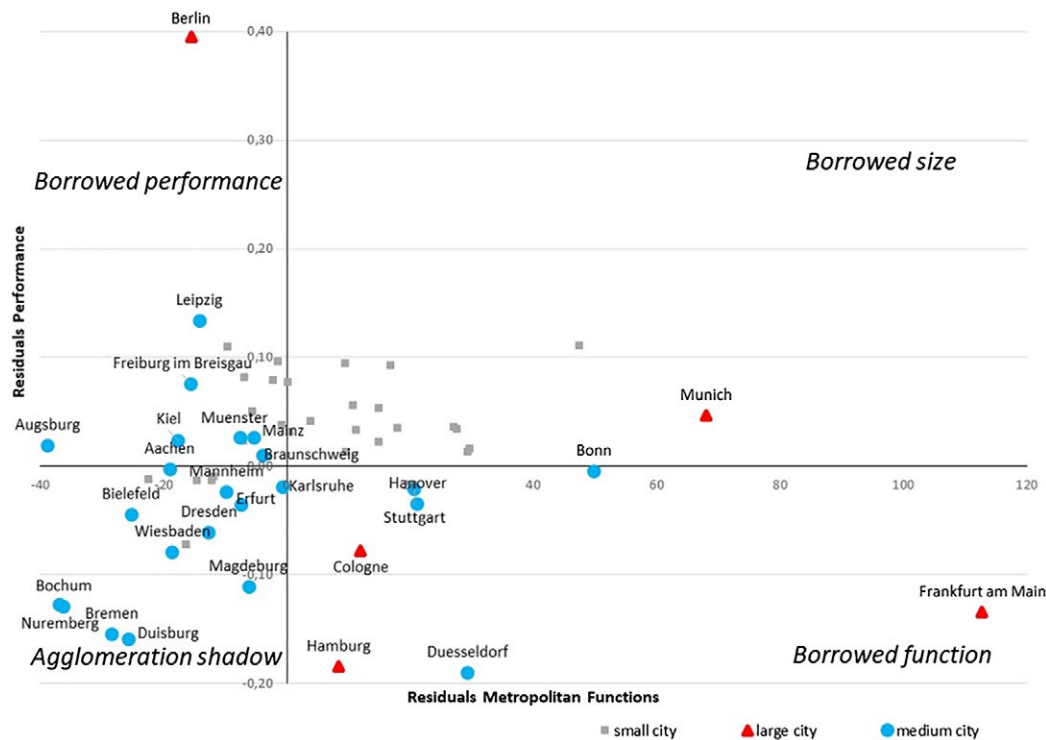
Figure 4. Employment growth in the cities of German urban regions (changes of employment shares 2008–2015).

cities in Germany. Despite this major development, there is no clear dominant growth in employment shares in specific urban regions. All functional urban regions host a mix of relatively low and high-performing hinterlands. For the more monocentric, high-performing regions, the picture is dominated by relatively well-performing regions. In addition to this general impression of changes in employment shares, our approach also shows the degree of functional or performance expectations derived from the regression residuals. Using the information gathered in the two regressions, we built the four-group classification listed in Table 2. We assigned a specific type of borrowed size to all cities with metropolitan functions.

The scatterplot in Figure 5 depicts the residuals of both regressions for the 60 municipalities in our study. Divided by the zero values of both axes and the three city-types large, medium and small city, the four dimensions

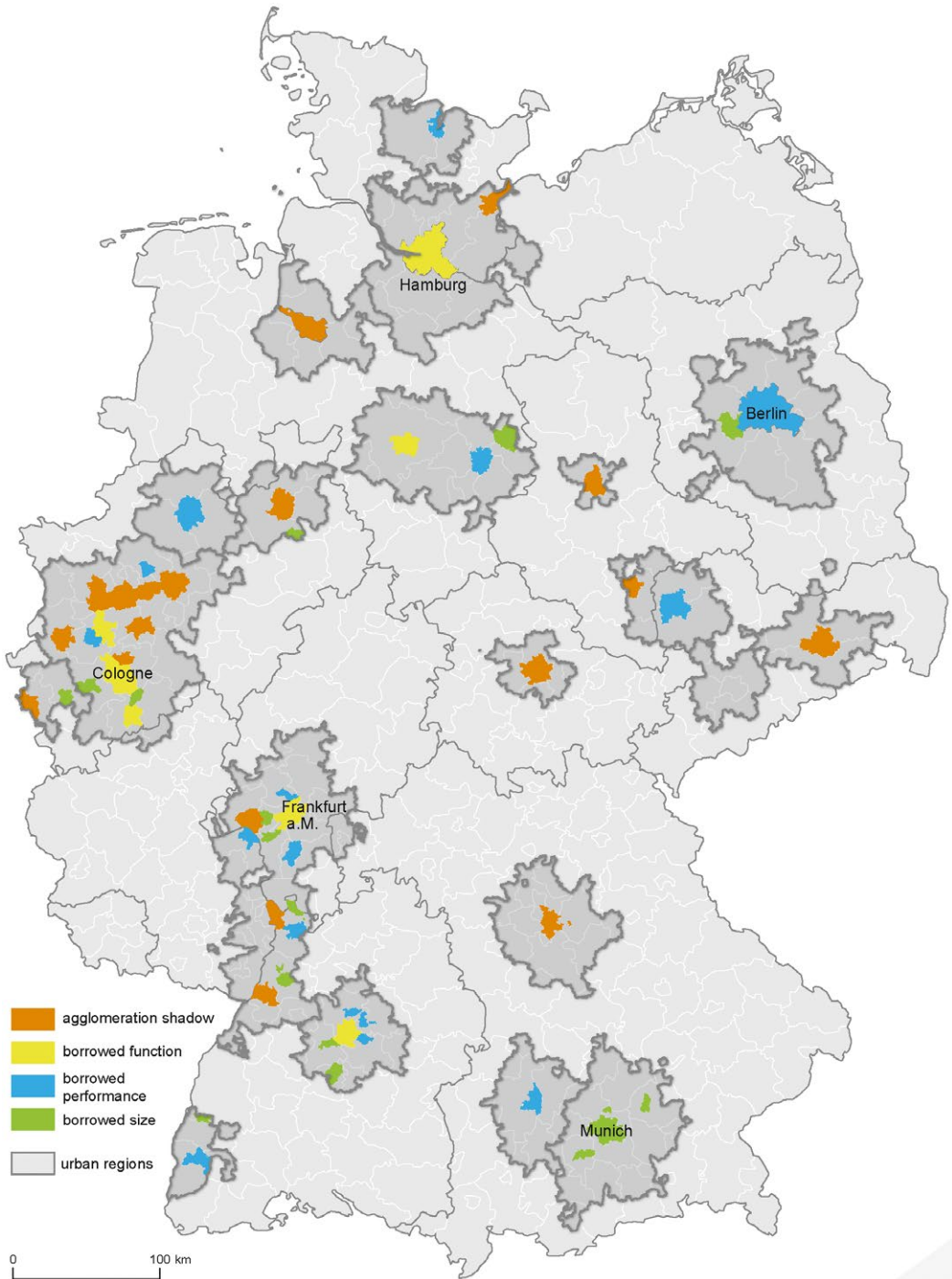
of the borrowed size concept can be differentiated. Most of the data pairs of residuals are scattered around the central of the axis, showing that there are detectable deviations from the expected values and that the data are not heavily influenced by outliers, which is another argument for the goodness of fit of our empirical approach. Only two data points are at the far ends of the scatterplot. One is Berlin, which performed much better than predicted while having slightly fewer metropolitan functions (why it is classified as ‘borrowing performance’). The other is the city of Frankfurt am main, which performed slightly worse than expected but hosts many more metropolitan functions given its size (why it is classified as ‘borrowing function’).

To extend the distribution of regression residuals, the classification results can be used to map the results for additional conclusions. Figure 6 and Table 3 show the borrowed size types of our sample of 60 municipalities.



Source: own calculations and illustration.

Figure 5. Scatterplot of regression residuals.



Source: own calculations. Cartography: Jutta Rönsch.

Figure 6. *Effects of borrowing size in German urban regions.*

Table 3. *Distribution of borrowed size types by city type.*

Borrowed size type	Large city	Medium city	Small city	Total
Agglomeration shadow	-	15	5	20
Borrowed function	3	4	-	7
Borrowed performance	1	7	8	16
Borrowed size	1	-	16	17
total	5	26	29	60

Source: own calculations and illustration.

Mapping the spatial distribution of the types and the relation with other cities that belong to a common functional urban region provides further insight into the (functional) relationship among larger, medium and small cities in their specific regional contexts.

The large cities of Berlin, Munich, Hamburg, Cologne and Frankfurt act as the leading centres of growth in the German urban system, showcasing a diverging picture in their individual borrowed size classification. Munich is the only city with above-average values in terms of metropolitanity and performance and therefore its development may be attributed to borrowing size from its hinterland. In addition, we find the dimension of borrowed size in 16 small cities (see Table 3) surrounding large cities such as Munich, Frankfurt am main, Cologne and Berlin. This indicates that small cities nearby larger agglomerations are successful in profiting from economic performance and functions due to their city-regional location. Thus these results correspond to Meijers and Burger (2017) assumption, that these processes occur more often in polycentric regions with equally sized neighbouring cities but is not reserved to small cities – in this case Munich.

Meijers and Burger (2017) assume that borrowing functions preferentially occur more often in large cities. This can be seen in seven cases: Hamburg, Cologne and Frankfurt am main, but also for the medium core cities Hanover, Dusseldorf, Bonn and Stuttgart. Berlin is the only large city with borrowing performance effects. All other large cities have a greater degree of metropolitanity than would be expected given their size. Therefore, large cities occupy an important position in the German urban system, supported by individual, nearby medium and small cities within a city-regional network. Some small and medium

cities (Esslingen, Ludwigsburg, Waiblingen near Stuttgart and Darmstadt, Mainz, Bad Homburg near Frankfurt am main and Neuss near Dusseldorf) in the surrounding area benefit from employment effects but do not fulfil any important metropolitan functions. As part of a polycentric urban region, they have access to the metropolitan functions of the neighbouring large city and agglomeration advantages (e.g. high-order services, institutions, and infrastructure such as research labs, educational institutions, and airports), resulting in positive structural effects. Other small cities in close proximity to large growing cities benefit from spatial proximity and connectivity to functions, thereby enriching themselves functionally and structurally (borrowed-sized effects). These regions are complemented by solitary urban regions Augsburg, Freiburg, Erfurt and Muenster. The coexistence of large, medium and small cities and the diverse spatial structure in urban regions is a key characteristic in the German urban system.

In contrast to the positive feedback and spillover effects 20 cities are characterised by negative effects from being located in close proximity to larger cities. They perform worse than expected and have fewer metropolitan functions given their size: they are affected by agglomeration shadow. These effects occur in two city-regional settings: in the structurally weak Ruhr area and Bergisches Land (Dortmund, Essen, Duisburg, Bochum, Wuppertal, Monchengladbach) in the polycentric Rhine-Ruhr urban region and in less-central monocentric urban regions defined by medium cores Dresden, Aachen, Bremen, Bielefeld, Magdeburg as well as the slightly polycentric urban regions Erfurt, Nuremberg.

In the first case, the large centres Dusseldorf, Cologne and Bonn cast an

Table 4. *Distribution of borrowed size types by spatial structure of the urban region.*

	City-region	Agglomeration shadow	Borrowed function	Borrowed performance	Borrowed size	Total
Monocentric urban regions	Aachen	1			1	2
	Augsburg			1		1
	Bielefeld	1				1
	Bremen	1				1
	Dresden	1				1
	Freiburg			1	1	2
	Kiel			1		1
	Leipzig			1		1
	Magdeburg	1				1
	Muenster			1		1
	Chemnitz					
Polycentric urban regions	Monocentric urban regions	5	-	5	2	12
	Berlin			1	1	2
	Erfurt	2				2
	Hamburg	1	1			2
	Hannover		1	1	1	3
	Munich				4	4
	Nuremberg	1				1
	Rhine-Ruhr	8	3	2	3	16
	RhineMain/ RhineNeckar	3	1	4	4	12
	/Karlsruhe					
	Stuttgart		1	3	2	6
	Polycentric urban regions	15	7	11	15	48

Source: own calculations and illustration.

agglomeration shadow over the Ruhr cities. In the second case, this cannot explain the classification of the more solitary urban regions. Size and influence of smaller German urban regions appear to be too small, as they are able to absorb a minimum level of metropolitan functions. This idea of a minimum threshold of 'urbanity' can be transferred to the whole urban system when examining the spatial distribution of borrowed size effects. Dominating urban regions such as Berlin, Munich or Hamburg cast an agglomeration shadow over the whole urban system in the sense of defining a minimum level of relative metropolitanity and economic performance for other urban region cores. If small and medium cities are more isolated in terms of the distance to major urban regions or the lack of supporting cities in their functional regions, it is more likely that this region will not be able

to keep up with the performance of the whole urban system.

Also, the German urban system allows for an analysis of different types of urban regions - monocentric or polycentric urban regions. Monocentric regions are regions with only one core city and polycentric urban regions are regions with more than two core cities (Table 4). Cities in large polycentric urban regions tend to be of the type 'borrowing functions'. So, these types of regions are able to transfer functional advantages to the entire urban region. By contrast, rather small monocentric urban regions without a major city in terms of population size do not generate any borrowed size effects. Importantly, larger monocentric regions are more effective in terms of borrowing performance, which is linked to the classical understanding of agglomeration economies.

CONCLUSION

Using German urban regions as an example, our paper contributes to the discussion on borrowing size effects. According to this concept, smaller cities that are part of larger functional urban regions can utilise high-order urban functions and economic dependencies, thereby boosting their regional performance. This is conceptualised in the four-dimension concept of Meijers and Burger (2017), which we further operationalised and assessed in terms of performance and function (see third section). We employed our empirical approach to classify cities along the four dimensions of the borrowed size concept to describe and evaluate the spatial distribution of borrowed size effects in the German urban system (see fourth section).

The German urban regions offer the opportunity to test the variety of the geography of borrowing size on local scale. Concerning the interpretation and understanding of our results, we state that urban regions are central and important driving forces in regional economics (Scott *et al.* 2001). The interrelations between large cities as international nodes and inner-city as well as suburban functional differentiations form an integrated system of spatial division of labour (Kloosterman & Musterd 2001; Parr 2014). In its geographical scope, the expansion and direction of regional urban development unfolds increasingly heterogeneous (Dijkstra *et al.* 2013). As we show in this paper, functional and economic growth is not solely concentrated on major cities. But the size of a city seems to be one central factor explaining the presence of metropolitan functions. Nevertheless, this general tendency needs to be confronted with the specific city regional contexts like the spatial structure. Whether a city is relatively small or large or is located in a monocentric or polycentric urban region has a significant impact on the four occurring borrowed size effects.

We provide evidence for economic and functional interrelations linked to large cities and their hinterland, which have been characterized as ‘interplaces’ by Phelps (2017). His argumentation is based on the fact that the contemporary economy is not purely divided into agglomerations (place) and networks

(space) but consists also of enclaves of export processing zones and arenas of trade fairs – which are labelled interplaces. Other concepts about polycentrism (Burger & Meijers 2012; van Meeteren *et al.* 2016; Rauhut 2017), Edge Cities (Garreau 1992) or New economic sub-centres (Krehl 2015; Kane *et al.* 2018) see urban regions as economic and social entities in which processes of concentration and de-concentration take place.

Those processes of concentration and de-concentration depend on internal and external agglomeration economies, which occur within functional urban regions and differ in size and functional structure – which we described with the four dimensions of the borrowed size concept. This interpretation is linked to the theoretical argumentation that agglomeration economies are not limited to the physical boundaries of a city. Moreover, they spill over to surrounding areas or cities (Parr 2002). This understanding is thereby enriched by the concept of borrowed size, in which the role of urban benefits is an essential explanatory variable for urban development (Burger *et al.* 2015; Hesse 2016; Malý 2016; Meijers & Burger 2017). To support regional development, agglomeration economies can be linked to the regional spatial and functional structure: improvements in regional transport and information and communication technologies support borrowing size in urban regions (Brezzi & Veneri 2015).

Our approach can be viewed as a starting point for further research employing the operationalised concept of borrowed size. Future research could focus on a differentiation of the metropolitan index on a much broader variety of urban functions to exploit the richness of the data and gather more detailed information on the typical functions for each borrowed function dimension, how this is linked to spatial co-agglomeration and if change interdependencies occur over time. It could be possible, that the distribution and specialisation of several metropolitan functions between cities in urban regions gives an indication of which metropolitan functions can be borrowed from surrounding smaller and medium cities.

In addition to developing the conceptual approach with other variables there is a need for a deeper understanding of the driving

forces of the spatial distribution of the borrowing size dimensions. This would explain the influencing factors for the ability of some cities to lend and use the critical mass or city size close to the large cities. Empirical indicators that would help explaining this relation are income indicators, transport connections and accessibility (see also Agnoletti *et al.* 2015) as well as the commuter flows within urban regions. For regional policies and policy measures, it is important to be able to identify and classify the phase and spatial shape of regional developments to adapt the right kind of urban policy (Dijkstra *et al.* 2013). In the future, small and medium cities in the surrounding should play a specific role for urban re-development and planning policies (Hesse & Siedentop 2018).

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